

Python full-stack course

About the program

Learn full-stack Python development with expert mentoring, covering modern web technologies across the frontend, backend, and database systems.

Key features

Industry-ready learning modules

24 weeks of learning

Review of progress by professionals working in the industry

Project-based learning

Course Curriculum

- HTML
- CSS
- Javascript
- React
- Python
- Django
- Postgresql
- Git & GitHub

Basic web technologies

HTML

- What is HTML
- Structure of an HTML document
- HTML elements and tags
- Headings, paragraphs, line breaks
- Links and images
- Lists (ordered, unordered)
- Tables
- Forms and input types
- Semantic HTML (header, footer, section, article)
- Audio and video elements
- Iframes
- Accessibility basics (alt, label, aria)

CSS

- What is CSS
- Inline, internal, and external CSS
- Selectors and properties
- Colors, units, and measurements
- Box model (margin, border, padding, content)
- Text and font styling
- Backgrounds and gradients
- Display and visibility
- Positioning (static, relative, absolute, fixed, sticky)
- Flexbox
- Grid layout
- Responsive design and media queries

Bootstrap

- What is Bootstrap
- Why use Bootstrap
- Bootstrap installation methods
- Bootstrap project structure
- Bootstrap grid system
- Containers
- Rows and columns
- Responsive breakpoints
- Typography
- Colors and background utilities
- Spacing utilities (margin and padding)
- Display and visibility utilities
- Flex utilities
- Position utilities
- Buttons
- Button groups
- Alerts
- Badges
- Cards
- Images and image utilities
- Tables
- Forms
- Navbar
- Dropdowns

Javascript

- What is JavaScript
- How to display output in JS
- Variables and data types
- Operators
- Conditional statements
- Loops
- Functions
- String and number methods
- Pre-defined Functions
- User-defined functions
- Form validation
- Error handling

Front-end

React

A JavaScript-based library to build front-end apps. Here we are learning to build a front-end with React

- What is React
- Why use React
- React environment setup
- JSX syntax
- Components and component structure
- Functional components
- Props
- State
- Handling events
- Conditional rendering
- Lists and keys
- Forms in React
- Controlled components

- Component lifecycle concepts
- Passing data between components
- React Router basics
- Dynamic routing
- Context API
- API integration
- Styling in React

Back-end

Python Programming Basics

- What is Python
- Python installation and setup
- Python syntax basics
- Variables
- Data types
- Type conversion
- Input and output
- Operators
- Conditional statements
- Loops
- Break and continue
- Functions
- Lambda functions
- Built-in functions
- Strings and string methods
- Lists and list methods
- Tuples
- Sets and set operations
- Dictionaries and dictionary methods
- Modules
- Import statements
- Virtual environments

- Exception handling
- Object-oriented programming concepts
- Classes and objects
- Constructors
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Decorators

Django

- What is Django
- Why use Django
- Django installation and setup
- Creating a Django project
- Creating Django apps
- Django app configurations
- Project structure overview
- Understanding MVT architecture
- URL routing
- Views and view types
- Function-based views
- Class-based views
- Templates and template syntax
- Template inheritance
- Static files handling
- Media files handling
- Creating Models
- Models and ORM basics
- Configuring the database
- Model fields and relationships
- Migrations
- Django admin panel
- Customizing admin

- Handling GET and POST requests
- CRUD operations
- Django settings and configuration
- Managing multiple apps
- Third-party package integration
- Authentication system
- User login and logout
- User registration
- Authorization and permissions
- Search and filtering
- Email integration
- Payment gateway integration
- Security basics (CSRF, XSS, SQL injection)
- Environment variables
- Debugging and logging

Django REST

- What is REST
- What is an API
- HTTP methods (GET, POST, PUT, PATCH, DELETE)
- What is Django REST Framework
- DRF installation and setup
- Serializers and serializer types
- URL routing in DRF
- CRUD operations using API
- Token authentication
- JWT authentication

Python Flask

- Introduction to Flask
- Defining models and relationships
- Manage CRUD operations
- Managing Database
- Flask project with CRUD operations

Database

Postgresql

An SQL-type database to build reliable data systems for web apps and APIs.

- What is PostgreSQL
- Database basics
- Relational database concepts
- Installing PostgreSQL
- PostgreSQL tools (psql, pgAdmin)
- Creating databases
- Connecting to a database
- Creating tables
- Data types in PostgreSQL
- Inserting data
- Selecting data
- Updating data
- Deleting data
- Constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL)
- Postgresql with Django

Django+React Integration

- Overview of Django and React integration
- When to use Django with React
- Frontend vs backend separation
- Setting up Django as backend
- Setting up React as frontend
- Project folder structure
- REST API concepts
- Using Django REST Framework with React

- Creating APIs in Django
- Serializers for React integration
- Consuming APIs in React
- Authentication flow overview
- Token-based authentication
- JWT authentication flow
- Login and signup integration
- Protected routes in React
- Role-based access control
- Handling permissions from backend
- State management basics
- Context API with Django APIs
- Error handling between frontend and backend
- Form submission from React to Django
- File uploads from React
- Environment variables setup
- API security best practices
- Serving React build with Django

AI integration

- Introduction to AI-based workflow
- Traditional workflow vs AI workflow
- Dealing with API's of AI
- Role of data in AI workflow
- Building an AI-based project

Git-Github

Get hands-on experience with modern version control systems that help you track project progress, manage updates, and work confidently in real-world development environments.

Learns git commands and GitHub workflow

- What is version control
- What is Git
- Git installation and setup
- Git configuration (username, email)
- Initializing a Git repository
- Git workflow basics
- Tracking files
- Git status
- Git add
- Git commit
- Git log
- Git ignore (.gitignore)
- Branches and branching concepts
- Creating and switching branches
- Merging branches
- Remote repositories
- What is GitHub
- Creating a GitHub account
- Creating a GitHub repository
- Connecting local repo to GitHub
- Push and pull operations
- Clone repositories
- Forking repositories
- Pull requests